

ArXAIv

An AI Generated Research Universe

Background

- Rise of reliance LLMs in generating paper content & reviews.
- Rise of VLM usage in scientific figure generation/validation.
- Misuse in science results in poor research.

Related Work:

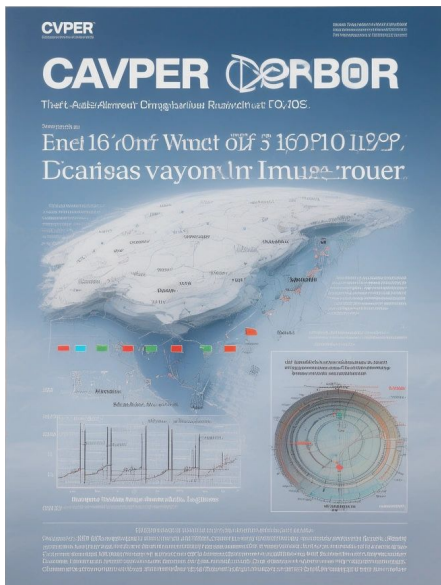
- <https://www.nature.com/articles/d41586-025-03506-6>
- <https://link.springer.com/article/10.1186/s41073-025-00165-z>
- <https://www.sciencedirect.com/science/article/pii/S2666389923000430>

Purpose

- The project raises questions like:
 - What makes a paper feel “real” or “credible”?
 - How does academic visual language signal legitimacy?
 - Could an entire fake canon of research be convincing?
 - What happens when we treat knowledge as a style, and remix it?

Method

Step 1) Try to use several SOTA VLMs, identify best one for creating realistic CV papers:



Experimentation of 2 Methodology

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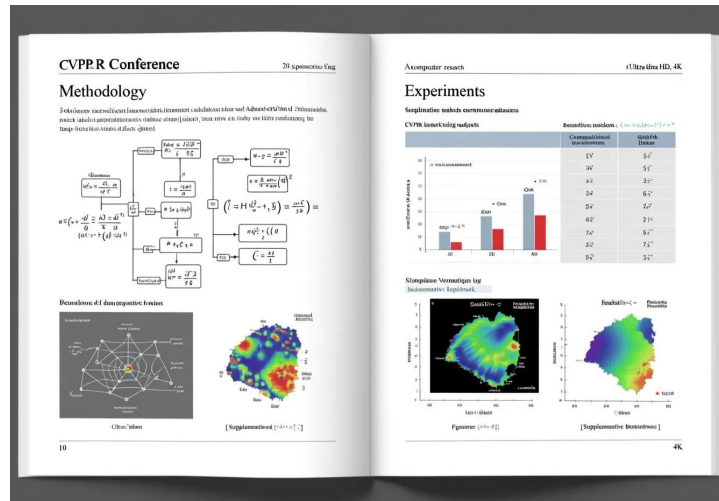
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► **Paper:** fraud mitigation

- [illegible]



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Stable Diffusion 3.5 Large

Qwen Image

More baselines:

CVPR CONFERENCE PAPER

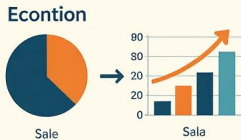
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Methodlogy

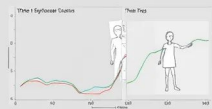
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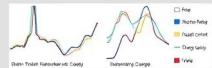
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3 Methodology

To ulopt a base n problem: fo learn to map input image x to ourput image y , given a training set with pairs (x_e, y_e) . To train its network, a wins a neural network structure of encoder f_{C_e} encodes an input image x into a latent representation module T . In the transformation module T operates and latent representation to produce $T(f_{C_e}(x_e))$, and the decoder converts to the output image $f_D(T(f_{C_e}(x_e)))$.

To train our transformantable module istce, I. trial Figure 1. In linear, a significane objective to minimized using paired training data, denoted is.

$$L = \sum_{e=1}^N L(f_D(T(f_{C_e}(x_e)), y_e)) \quad (1)$$

Figure 1: Illustration of the proposed nework architecture.

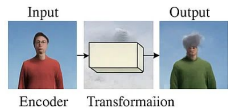


Figure 1: Illustration of the proposematr-

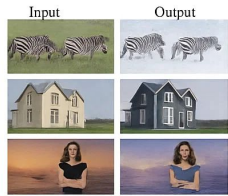


Figure 2: Example results of our approach.

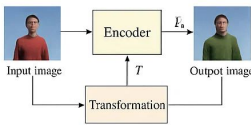


Figure 1: Illustration of the proposed network architecture.

4 Experiments

We evaluate using the proposed approach on a dataset.

4.1 Quantitative Results

Table 1 presents resus fly results comparison our method with put baselines. In tems of mean squared error (MSE)) and structural similarity index,

4.2 Qualitative Results

Figure 2 provide an examples of generated outputs ly.

Table 1: Quantitative comparison with baseline methods.

Method	MSE	SSIM
Baseline A	0.120	0.65
Baseline B	0.085	0.78
Ours	0.057	0.83

2 Example results of our approach.

FIGURE: Example results of our approach.

Figure	Input	Output
A	0.120	0.65
B	0.085	0.78
Ours	0.057	0.83

Hi-Dream-L1-Fast

Flux 1 Schnell

SORA

Method

Step 2) Identify key evaluation metrics:

1. **Reference Authenticity Index:** Ratio of real vs hallucinated references
2. **Institutional Hallucination Score:** % of academic institutions that do not exist + author metrics (most common names)
 - a. Consider the most frequently cited institutions as well
3. **Visual Convention Similarity:** How closely AI figure descriptions match real CV paper figures
 - a. CLIP model similarity checker
4. **Title Similarity:** How closely AI paper titles resemble real papers
 - a. Cluster titles by subtopic as well; automate by running through a program to find 'keywords'
 - b. Find most similar titles manually, then get cosine similarity
5. **Embeddings:**
 - a. Map for paper content embeddings, identify key clusters
 - b. Map for figure embeddings, identify key clusters

Experiments

- Used SORA to generate 100 AI generated papers: [Papers](#)
- Separate folder for AI-figures: [AI Figures](#)
- Manual parsing for author names, title, and affiliation.

GitHub Repository for Data Extraction: <https://github.com/allisonandreyev/arxaiv>

3D Force Graph: <https://github.com/vasturiano/3d-force-graph>; use to generate embedding graph.

- Paper text extracted using OCR was cleaned of typo/mistext using GPT 5.1 prior to generating embeddings.

Results so far:

Cluster	▼	%	▼
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Object detection		21	
Semantics/segmentation		25	
Image generation		12	
Localization/Spatiotemporal		18	
Visual Features/networks		24	

Features	▼	% total	▼
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All features present		69	
Missing title		19	
Missing affiliations		7	
Missing References		8	

- Papers are clustered around a few key topics in computer vision research.
- Many contain missing titles or wrongly generated affiliations, but content and scientific figures remain intact and consistent.

Affiliations Map

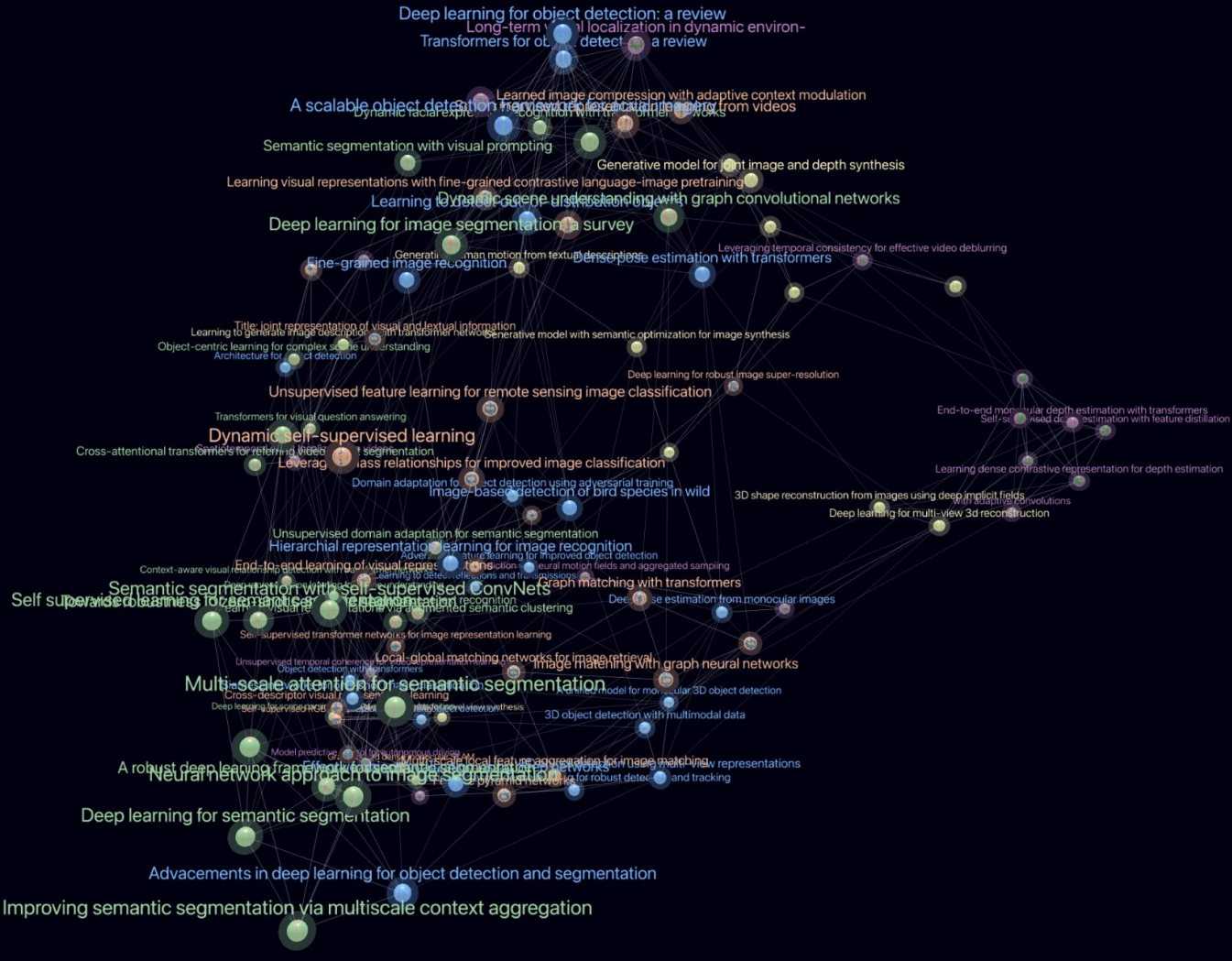
Raw # ▾	Affiliation ▾	% Total Affiliations ▾	% Total Authors ▾
107	Hallucinated	33.75394322	31.47058824
77	Mutated	24.29022082	22.64705882
44	Stanford	13.88012618	12.94117647
23	None	7.255520505	6.764705882
23	U Toronto	7.255520505	6.764705882
19	MIT	5.993690852	5.588235294

Top 4 Affiliations

- Hallucinated – completely made up.
- Mutated – resemble real life institution or business, but a few characters deviate.

Observations:

- Most affiliations are from top computer science research universities, though a large percentage of text is still mutated.
- Most papers are from American researchers/institutions, likely a reflection of training data.

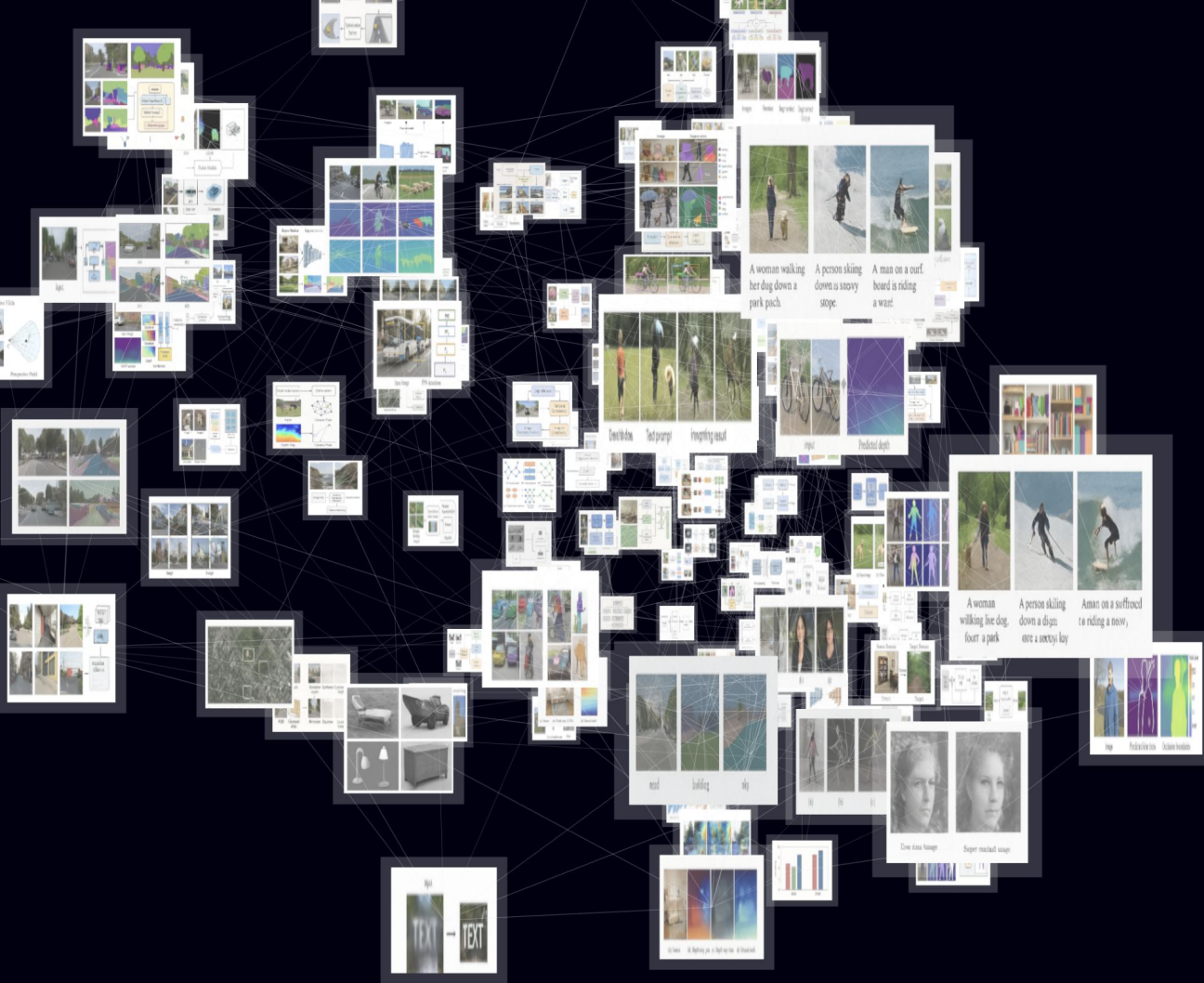


Paper Content Embedding Map

- Paper content remains roughly relevant to the title (clusters based on titles form distinct groups in embedding map)
- Content overall remains similar amongst colors, with few clusters.

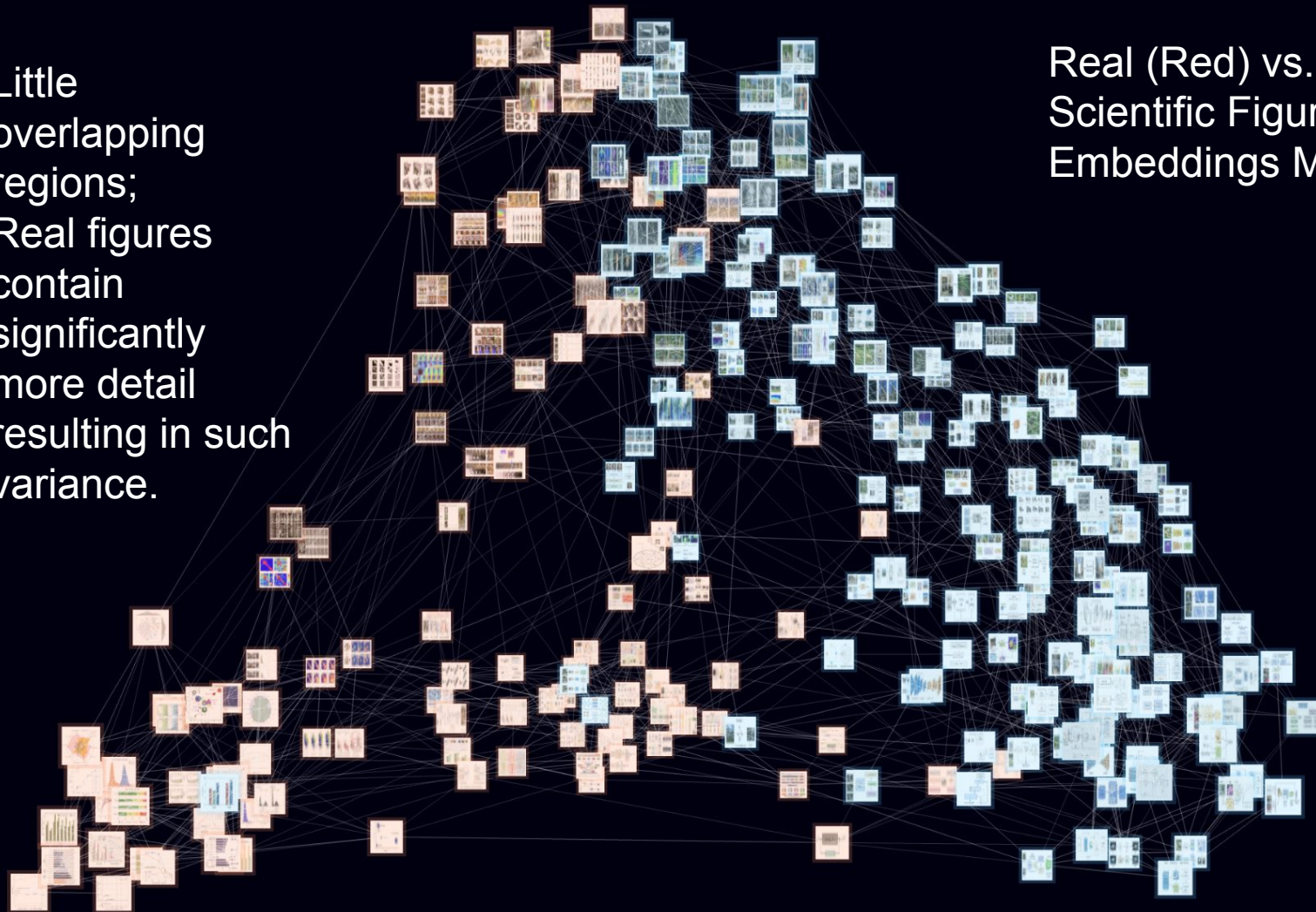
AI Generated Scientific Figure Embedding Map

- Figure designs have 2 key structures: table/graph based, and image based.
- Little design variance.



- Little overlapping regions;
- Real figures contain significantly more detail resulting in such variance.

Real (Red) vs. Blue (AI)
Scientific Figure
Embeddings Map



Next Steps (Currently in progress)

- Generate video and web page simulating results.
- Design more baselines/experimentation.
- Publish dataset as hallucinated/flawed scientific figure collection.